Fiscal Year:	FY 2022	**Task Last Updated:**	FY 03/11/2022
PI Name:	Koochesfahani, Manoochehr Ph.D.		
Project Title:	Phase 2: ZBOT-2 Thermometry Diagnostics Development		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Fluid physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	Physical Sciences Unsolicited
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No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Olson, David A Ph.D. (Michigan State University)		
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In-orbit or on-surface cryogenic storage of propellant and life support consumables is essential to all stages of NASA's Exploration Program. However, heat leakage from the surroundings results in liquid vaporization and significant mass loss and/or significant increase in the storage tank pressure. To mitigate the unwanted effects of cryogen vaporization, the Zero Boil-Off (ZBO) concept is being considered as an innovative approach for ventless means of controlling tank pressure through the application of active heat removal and forced liquid jet mixing. The Zero Boil-Off Tank (ZBOT) experiment on the International Space Station (ISS) is designed to examine the ZBO concept and, along with computational fluid dynamics (CFD) modeling efforts, provide information on the fundamental physics of the underlying complex thermofluids environment, which is needed for engineering design optimization of the ZBO concept. The optical diagnostics for whole-field thermometry that are being developed in the current proposal is crucial to providing scientific data and measurements for validation and further refinement of the two-phase CFD models developed by the ZBOT microgravity investigation. The ZBOT project directly aligns with NASA exploration needs for

Task Description:	Cryogenic Fluid Management of future propellant and life support storage tanks, and the ZBO tank pressure control is an important technology achievement that enables human missions to Moon, Mars, and other planets. Under NASA support we have successfully demonstrated the feasibility of a novel non-intrusive optical method for whole-field temperature measurement suitable for the ZBOT microgravity experiment. The technique of Quantum Dot Thermometry (QDT), specifically developed for the planned ZBOT experiments on the ISS, relies on laser induced fluorescence (LIF) thermometry based on quantum dots (QD) that are solubilized/dispersed in the working fluid of the ZBOT experiment. We have quantified the influence of the various interconnected parameters, and the tradeoffs among them, that influence the ability to make accurate temperature measurements, and we have shown that the science requirements are expected to be met by the judicious selection of detection strategy and data processing methods. The objectives of the proposed research are to capitalize on our progress to date and further develop the technique with a focus on the tradeoffs connected to the actual engineering constraints in implementation and design optimization of experiments on the ISS. A core element of this effort will be the close collaboration with NASA engineering and hardware design experts, and providing our scientific know-how in fluid flow optical diagnostics and data processing/analysis, as we transition the measurement technique to actual implementation for the ZBOT experiments on the ISS.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The ZBOT project directly aligns with NASA exploration needs for Cryogenic Fluid Management of future propellant and life support storage tanks, and the ZBO tank pressure control is an important technology achievement that enables human missions to Moon, Mars, and other planets. Development of Quantum Dot Thermometry enables the measurements needed for obtaining scientific data for validation and refinement of the two-phase two-component CFD models being developed as part of the ZBOT project.
Task Progress:	NOTE: Follow-on project to "ZBOT-2 Thermometry Diagnostics Development," grant 80NSSC19M0128, with same Principal Investigator (PI), Dr. Manoochehr Koochesfahani. Over the last several years under NASA support, we have developed the novel optical method of Quantum Dot Thermometry (QDT) for non-intrusive whole-field mapping of the temperature field in the ZBOT microgravity experiment. The technique relies on ratiometric laser induced fluorescence (LIF) thermometry, based on quantum dots (QD) that are solubilized/dispersed in the working fluid of the Zero Boil-Off Tank (ZBOT) experiment, Perfluoro-n-Pentane (PnP). The objectives of the current research effort are to capitalize on our progress to date and further develop the QDT technique for the ZBOT-NC experiment (Ed. Note: "NC" refers to the addition of a non-condensable gas in the tank) with a focus on the tradeoffs connected to the actual engineering constraints in implementation and optimization of experiments aboard the International Space Station (ISS). A core element of this effort is the close collaboration between our scientific know-how in fluid flow optical diagnostics, data processing/analysis, and the NASA engineering and hardware design experts, as we transition the measurement technique to actual implementation for the ZBOT-NC experiments aboard the ISS. In the early part of this year, the QDT development faced a new challenge in that the QDs we have been using for several years are no longer sold by the supplier. After a concerted effort to identify alternate QDs that can be solubilized/dispersed in PnP, and are suitable for QDT, we have determined that the PnP-soluble QDs developed for us by NanoOptical Materials (NOM) are viable replacements for the original QDs. The optical properties of the new QDs have been characterized, along with their expected QDT performance curves for selected detection strategies. In our previous reports and presentations, we have described that the two-camera approach provides the best performance in ratiometric thermometry in terms of both accuracy and spatial resolution. A single color camera with a Bayer filter sacrifices both spatial resolution and, more importantly, the Bayer filter spectrum leads to reduced ratiometric temperature response, and therefore, less accuracy. Due to several implementation constraints for ISS experiments, however, a single color camera is the preferred detection strategy of QDT for ZBOT-NC. An effort was launched to consider the possibility of improving the QDT performance of the single color camera with Bayer filter by replacing it with a multi-chip camera. A multi-chip camera is essentially a two-camera system in a compact packaging that contains the beam splitting optics, the two image detectors, and their color filters inside a body with a footprint comparable to the single color-camera, which would be suitable for the constraints for ZBOT-NC. We selected a commercially available multi-chip camera for evaluation because its off-the-shelf color filter set was estimated to produce (and later measured) nearly a factor of two improvement in QDT temperature sensitivity compared to the color camera already in use. While the improvement in temperature sensitivity directly maps into commensurate improvement in QDT temperature measurement uncertainty, the camera's complementary metal oxide semiconductor (CMOS) sensors performed significantly worse than the sCMOS sensor of the single color camera in terms of signal-to-noise ratio. The net effect was the multi-chip camera performed worse in terms of temperature uncertainty compared to the single color camera, and therefore, not considered a viable option for ZBOT-NC experiments. One challenge in defining the science requirements for the ZBOT-NC experiments is to determine whether a specified pair of temperature measurement uncertainty and spatial resolution can reveal the changes in the thermal field in microgravity due to the addition of non-condensable gas and, therefore, the changes in the underlying flow physics. The requirements for whole-field temperature data to reveal such flow physics are not the same as those typically specified for single-point sensors. We feel a great deal of progress can be made towards finalizing the science requirements for QDT by taking advantage of the available computational data on the effect of non-condensable gas in the ZBOT thermal-flow field. We used the computational data sets from Dr. Kassemi's Computational Fluid Dynamics (CFD) group to analyze the implications of several of the known sources of noise and data corruption on the ability of QDT to meet the overall objectives of ZBOT-NC. [Ed. Note: Dr. Mohammad Kassemi is the PI for "The Zero Boil-Off Tank-NonCondensibles (ZBOT-NC) Experiment"]. The factors studied were the size of the spatial averaging domain, the random uncertainty due to camera performance, streaky patterns generated by the ZBOT tank and potentially the flow itself, and regions of missing data due to obstructed regions in the tank such as those caused by the heating tape, and other imaging artifacts. Simulating these non-ideal realities of experimental data demonstrated to the science team the current state of QDT and its proximity to meeting the science objectives. It also highlighted the critical need to assess the QDT performance curves in the actual flight hardware and its benchmarking against actual ground-based experiments. The results and

	conclusions have been shared with NASA / ZIN Technologies and have expedited the scheduling of the needed experiments just described.
Bibliography Type:	Description: (Last Updated: 03/02/2026)
Abstracts for Journals and Proceedings	Olson D, Koochesfahani M. "Quantum dot thermometry optimization for the ZBOT-NC experiment." 37th Annual Meeting of the American Society for Gravitational and Space Research, Baltimore, MD, November 3-6, 2021. Abstracts. 37th Annual Meeting of the American Society for Gravitational and Space Research, Baltimore, MD, November 3-6, 2021. , Nov-2021